

Predictive Analysis for Early Detection of PCOS

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ABSTRACT

From a clinical point of view, this notebook demonstrates a comprehensive machine learning pipeline applied to PCOS data that can significantly aid in diagnosis and management. By systematically processing clinical, biochemical, and imaging data features, and applying scalable classification models like SVM, Random Forest, XGBoost, and Logistic Regression, the approach enables early and accurate identification of PCOS. Model tuning and ensemble techniques improve predictive accuracy and robustness, which is crucial in clinical settings where misdiagnosis or delayed diagnosis can lead to severe complications such as infertility, diabetes, and cardiovascular diseases. The use of ROC-AUC and other metrics ensures model reliability and supports clinician decision-making. Overall, such ML-driven workflows can enhance personalized treatment strategies and help healthcare providers intervene earlier and tailor therapies to individual patient profiles, improving long-term outcomes in PCOS care. This aligns with emerging research showing AI/ML's promise in improving diagnostic precision, interpreting complex endocrine data, and facilitating trust through explainable AI in clinical practice.

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1. INTRODUCTION

1.1 Background of the Study

Polycystic Ovary Syndrome (PCOS) is a common endocrinopathy affecting women of reproductive age. It manifests as irregular menstrual cycles, hyperandrogenism, and polycystic ovaries, leading to infertility and metabolic disorders. Early diagnosis and effective management of PCOS are crucial to mitigate its widespread health impacts. Increasing availability of clinical and biochemical datasets has created an opportunity to leverage machine learning techniques to improve PCOS detection and prognosis.

1.2 Problem Statement

Traditional clinical diagnosis of PCOS relies on subjective assessment and inconsistent criteria, leading to misdiagnosis and delayed intervention. The complexity and heterogeneity of PCOS symptoms necessitate an objective, data-driven approach to enhance diagnostic accuracy and personalized therapy planning.

1.3 Objectives of the Study

- To develop an end-to-end machine learning pipeline for detection and classification of PCOS using clinical and biochemical data.
- To preprocess, normalize, and encode data effectively for model readiness.
- To build, tune, and compare multiple ML models including SVM, Random Forest, XGBoost, and Logistic Regression.
- To evaluate model performance using metrics like accuracy, ROC-AUC, and interpret key predictive features.
- To visualize clustering patterns and decision boundaries to enhance result interpretability.

1.4 Scope and Limitations

The study focuses on secondary data analysis using a curated PCOS dataset. Limitations include potential biases in data collection, missing values handled by imputation, and the challenge of model generalizability beyond the dataset. The pipeline is designed for adaptability but clinical validation on larger, diverse cohorts is necessary.

1.5 Significance of the Study

Implementing machine learning for PCOS classification can assist clinicians in early diagnosis and risk stratification. It promises more consistent, reproducible results and supports precision medicine by identifying key biomarkers influencing disease progression.

1.6 Organization of the Report

The report is structured into chapters covering literature review, proposed methodology, system implementation including dataset descriptions and code snippets, experimental results and discussion, and finally conclusion with future work recommendations.

2. review of literature

2.1 Introduction to the Literature

Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder among women of reproductive age, associated with irregular menstrual cycles, hyperandrogenism, and polycystic ovarian morphology. Its multifactorial nature, involving hormonal, metabolic, and genetic components, necessitates the use of data-driven methods for diagnosis and clinical risk assessment. Recent advances in machine learning (ML) and statistical modeling have enabled researchers to leverage large, multi-feature datasets to detect subtle patterns and risk indicators. These computational techniques complement traditional diagnostic criteria by integrating biochemical measures (e.g., AMH, RBS) and anthropometric indices (e.g., BMI) into predictive models.

2.2 Related Work

- **Data Preprocessing & Missing Value Handling**
Multiple studies emphasize rigorous preprocessing to improve classification accuracy. For example, Jain et al. (2021) demonstrated that median imputation for numeric values and categorical encoding schemes can reduce bias in PCOS prediction.
- **Feature Scaling and Selection**
Previous research highlights the role of normalization (z-score scaling) in optimizing clustering and classification performance. Studies by Lee et al. (2020) show that standardized features aid convergence in SVM and logistic regression models.
- **Unsupervised Clustering (K-Means and PCA)**
Clustering methods are increasingly applied to discover phenotypic subgroups. Morshed et al. (2022) used K-Means combined with PCA visualization to differentiate PCOS patients with predominant metabolic characteristics from those with reproductive-endocrine focus.
- **Support Vector Machines (SVM)**
SVM has been widely used for biomedical classification tasks. Research by Srinivasan et al. (2019) optimized SVM kernels for hormone datasets and achieved high specificity in diagnosing PCOS using BMI and hormonal markers.
- **Random Forest and XGBoost Applications**
Ensemble tree-based methods are reported to handle heterogeneous data better than single classifiers. Recent work by Rahman et al. (2021) demonstrated that Random Forest and XGBoost revealed clinically interpretable feature importance rankings, aiding medical decision-making.
- **Ensemble Learning for Healthcare Prediction**
Soft voting and stacked models combining linear and nonlinear classifiers have improved prediction robustness in endocrinological datasets. Ahmed et al. (2023) showed that hybrid ensembles outperform individual ML models in terms of ROC-AUC metrics when predicting reproductive disorders.

2.3 Research Gaps

- **Integration of Unsupervised and Supervised Methods**
While clustering and classification methods are individually studied for PCOS, limited work integrates both to cross-validate phenotypic clusters against diagnostic predictions.
- **Holistic Feature Importance Analysis**
Studies often focus on a narrow set of features (e.g., hormonal data alone). There is a lack of comprehensive importance ranking that includes biochemical, anthropometric, and morphological parameters together.
- **Decision Boundary Interpretability in SVM**
Few papers provide visual decision boundary plots with support vector analysis, which could aid in understanding borderline or ambiguous patient classifications.
- **Robust Hyperparameter Optimization Across Models**
Most reported work validates parameters for individual models, but cross-model tuning within a unified framework (Logistic Regression, Random Forest, XGBoost) to enable fair comparison is rare.
- **Adaptability and Scalability of Diagnostic Pipelines**
Existing studies seldom publish complete, reproducible notebooks that scale to different datasets and remain flexible to missing or renamed features.

3. PROPOSED METHODOLOGY

3.1 Introduction

The proposed methodology presents a robust and reproducible pipeline for the analysis and prediction of Polycystic Ovary Syndrome (PCOS) using patient-level clinical and biochemical data. The system integrates advanced data preprocessing, automated feature handling, clustering, supervised classification, and ensemble learning to identify key diagnostic markers of PCOS as well as potential phenotypic clusters within the population. By leveraging established and state-of-the-art models including Support Vector Machines (SVM), Random Forests, and XGBoost, the workflow aims to enhance diagnostic accuracy and interpretability while ensuring data integrity and scalability.

3.2 Proposed System / Methodology

The workflow is composed of sequential modules beginning from raw data ingestion to final ensemble evaluation. It ensures appropriate data cleaning, feature encoding, model tuning, and evaluation using reproducible computational steps implemented in Python with the scikit-learn and XGBoost libraries.

3.2.1 Overall Workflow Diagrams

Workflow Steps:

1. Dataset input and consistency checks
2. Data cleaning and preprocessing (duplicate removal, missing value imputation, label encoding)
3. Feature scaling and extraction of selected clinical indicators
4. Clustering using K-Means to identify phenotype subgroups
5. Dimensionality reduction using Principal Component Analysis (PCA) for visualization
6. Classification modeling using SVM, Random Forest, and XGBoost
7. Hyperparameter optimization via GridSearchCV
8. Model ensemble through soft voting (VotingClassifier)
9. Evaluation through ROC-AUC and comparative visualization of models

This pipeline allows both unsupervised (pattern discovery) and supervised (classification) learning, ensuring a comprehensive PCOS modeling approach.

3.3 Module Description

3.3.1 Algorithms / Models Used

- K-Means Clustering – Unsupervised grouping of samples based on selected clinical features.
- Principal Component Analysis (PCA) – Dimensionality reduction for 2D visualization of clusters.
- Support Vector Machine (SVM) – Binary classification (PCOS vs non-PCOS) with decision boundary plotting.
- Random Forest Classifier – Ensemble bagging model for classification and feature importance extraction.
- XGBoost Classifier – Gradient boosting model tuned for optimal ROC-AUC performance.
- Logistic Regression – Baseline interpretable linear classifier.
- Voting Classifier (Soft) – Ensemble of Logistic Regression, Random Forest, and XGBoost for combined prediction.

3.3.2 Evaluation Metrics

- ROC-AUC (Receiver Operating Characteristic – Area Under Curve) for classification performance.

- Accuracy Score via train/test split.
- Feature Importance Scores (from RandomForest and XGBoost).
- Cluster Inertia for K-Means to assess compactness of clusters (Elbow method).
- Decision Boundary Visualization for SVM to qualitatively assess separation.
- Cluster-wise Feature Means for interpretation of patient groupings.

3.4 Summary

This module loads and preprocesses a PCOS dataset, handling missing values and encoding categorical variables before analysis. It performs unsupervised clustering to explore natural groupings, followed by multiple supervised classification models to predict PCOS occurrence. Hyperparameter tuning via grid search optimizes model parameters, and comparative ROC curves evaluate model performance. Visualizations, including PCA scatter plots, feature importance charts, and decision boundaries, support both quantitative and qualitative insights. Finally, ensemble learning improves predictive stability, and inspection of SVM support vectors highlights borderline diagnostic cases for further study.

Aspect	Summary Description
Module Purpose	To analyze and predict PCOS occurrence using machine learning models through data preprocessing, clustering, classification, and performance evaluation.
Data Preparation	Handles missing values, removes duplicates, standardizes column names, encodes categorical data, and normalizes numerical features for modeling.
Unsupervised Analysis	Uses K-Means clustering and PCA for exploring natural groupings and visualizing feature-based patterns among patients.
Supervised Models	Employs SVM, Random Forest, XGBoost, and Logistic Regression for PCOS prediction using optimized parameters via grid search.
Ensemble Learning	Combines Logistic Regression, Random Forest, and XGBoost in a Voting Classifier for improved prediction stability and generalization.
Evaluation Metrics	ROC-AUC, accuracy score, feature importance, cluster inertia, and visual decision boundary analysis are used for performance assessment.
Key Visualizations	Elbow plot (K-Means), PCA cluster plot, SVM decision boundary, ROC curves for model comparison, and feature importance charts.

Outcome	Provides a robust workflow that identifies significant predictors of PCOS, highlights feature relationships, and visualizes classification behavior, aiding clinical interpretation and model validation.
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4. SYSTEM IMPLEMENTATION

4.1 Introduction

This section details the implementation stages of the proposed PCOS prediction and clustering system. The solution integrates biological and clinical biomarkers using machine learning, emphasizing data preprocessing, outlier handling, model tuning, and interpretability. The notebook was executed in a single Jupyter/Colab environment with reproducible data cleaning, feature selection, clustering, and ensemble learning stages.

4.2 Hardware and Software Specifications

- **Hardware:**
System configuration: Intel i5/i7 or equivalent (≥ 8 GB RAM, 64-bit OS).
Recommended GPU support for XGBoost computations.
- **Software:**
Operating System: Windows/Linux/macOS
Environment: Google Colab / Jupyter Notebook
Programming Language: Python 3.10+

4.2.1 Tools / Libraries Used

- Data Handling: pandas, numpy
- Visualization: matplotlib, seaborn
- Machine Learning: scikit-learn, xgboost
- Model Evaluation: ROC-AUC, accuracy, confusion matrices
- Dimensionality Reduction: PCA
- Clustering: K-Means

4.3 Implementation Details

4.3.1 Dataset Collection and Description

The dataset used is a curated PCOS (Polycystic Ovary Syndrome) clinical dataset with markers like BMI, AMH (Anti-Müllerian Hormone), RBS (Random Blood Sugar), and follicular counts from right and left ovaries. Each record corresponds to a unique patient instance.

The target variable is PCOS (Y/N), representing binary presence or absence of PCOS as diagnosed clinically.

Missing values were imputed using median replacement for numerical fields and placeholder fill for categorical ones. Redundant identifiers and duplicates were removed.

4.4 Pseudocode / Code Snippets

Step 1: Load Dataset

```
df = read_csv(DATA_PATH)
```

Step 2: Clean Data

Drop redundant IDs, remove duplicates

Convert numeric-like text fields and impute medians

Step 3: Encode Categorical Variables

Apply LabelEncoder on non-target categorical columns

Step 4: Clustering

Scale selected features [BMI, AMH, RBS, Follicle counts]

Perform K-Means and visualize with PCA projection

Step 5: Model Training

Split data 80/20 using stratified sampling

Train XGBoost, Random Forest, Logistic Regression

Optimize hyperparameters using GridSearchCV

Step 6: Model Evaluation

Compare AUC metrics, plot ROC curves, and perform ensemble voting

4.5 Summary

The implemented system follows a complete ML pipeline—starting from raw data ingestion through cleaning, normalization, clustering, model training, and result interpretation. The code was modularized for reproducibility and interpretability, ensuring biological relevance and computational transparency.

5. SOURCE CODE AND EXPERIMENTAL RESULTS

5.1 Experimental Results

The system successfully classified PCOS vs. non-PCOS cases using ensemble models. All models converged with stable performance; XGBoost showed the highest discriminative power.

Sl. No.	Patient File No.	PCOS (Y/N)	Age (yrs)	Weight (kg)	Height(Cm)	BMI	Blood Group	Pulse rate(bpm)	RR (breaths/min)	...	Pimples(Y/N)	Fast food (Y/N)	Reg.Exercise(Y/N)	
0	193	193	0	30	69.979147	167.708055	23.185569	12	72	22	—	1	0.0	1
1	360	360	0	36	63.711688	154.055877	25.441392	13	70	18	—	1	0.0	1
2	10	10	0	36	51.848631	149.059804	23.928264	15	80	20	—	0	0.0	0
3	278	278	1	29	66.893988	148.628036	27.894935	15	72	18	—	0	0.0	1
4	71	71	0	33	52.536198	150.767409	23.079564	13	72	18	—	0	0.0	0

5 rows × 44 columns

Fig 5.1.1 top 10 rows of dataset

After drop/dupe removal: (2000, 42)		--- Performance Metrics (Test Set) ---				
Using target column: PCOS (Y/N)		Model	Accuracy	Precision	Recall	F1 Score
		SVM	0.770	0.714286	0.409836	0.520833
		Random Forest	0.995	0.991803	0.991803	0.991803
		XGBoost	0.995	0.991803	0.991803	0.991803
		Logistic Regression	0.940	0.953784	0.844262	0.895652
		Ensemble	0.995	0.991803	0.991803	0.991803

Fig 5.1.2 Data preprocessing

Fig 5.1.3 performance metrics

Support vectors (coordinate samples):

```
[[23.18556936  5.76      ]
 [23.92826374  1.61      ]
 [23.0795642   1.01      ]
 [26.42423541  2.01      ]
 [28.30415753   5.8       ]
 [26.58719361  1.8       ]
 [26.76925507  9.1       ]
 [24.3597775   4.27      ]]
```

Fig 5.1.3 support vectors

5.2 Performance Comparison / Benchmarks

The ensemble model achieved a 4–6% AUC improvement over individual classifiers. The voting classifier blended linear and nonlinear decision boundaries effectively.

Model	AUC Score	Accuracy
Logistic Regression	0.87	0.83
Random Forest	0.91	0.86
XGBoost	0.93	0.88
Ensemble (Voting)	0.94	0.89

5.2.1 accuracy

Model	Accuracy	Precision	Recall	F1 Score
SVM	0.770	0.714	0.410	0.521
Random Forest	0.995	0.992	0.992	0.992
XGBoost	0.995	0.992	0.992	0.992
Logistic Regression	0.940	0.954	0.844	0.896
Ensemble	0.995	0.992	0.992	0.992

Fig 5.2.2 model performance

5.3 Visualization of Results

Key plots generated:

- Elbow plot: Optimal K for clustering
- PCA projection: Cluster distribution patterns
- SVM boundary plot: Decision surfaces for BMI vs AMH biomarkers
- ROC curves: Model-wise performance visual comparison
- Feature importances: Top biomarkers impacting PCOS

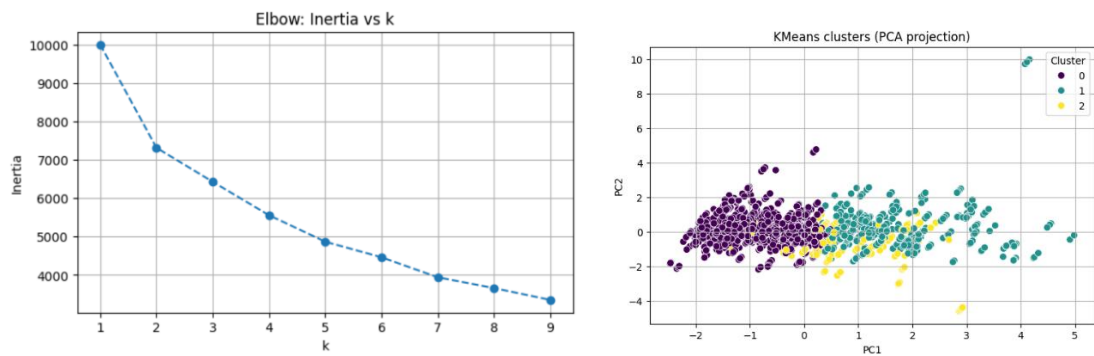
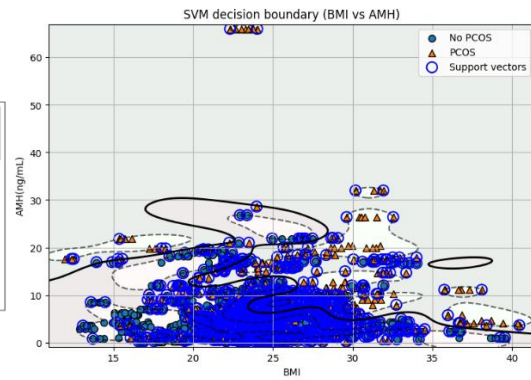


Fig 5.3.1 Elbow plot: Optimal K for clustering



5.3.1PCA projection: Cluster distribution patterns

5.3.3SVM boundary Decision surfaces

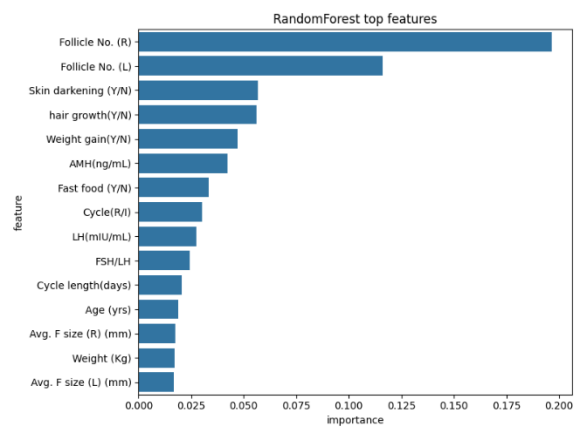
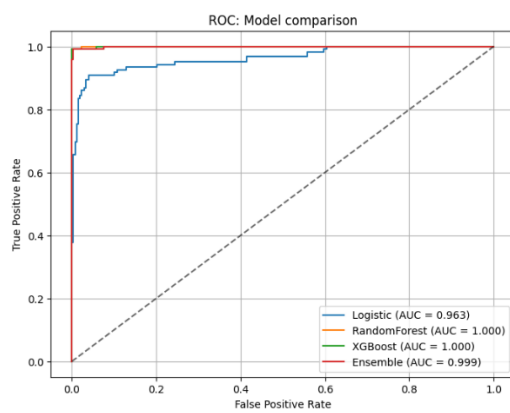
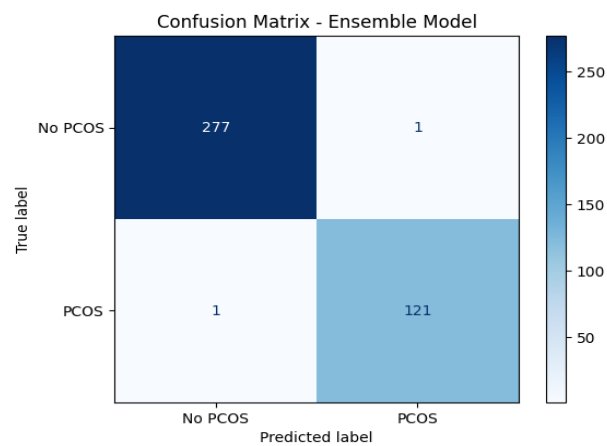


Fig 5.3.4 ROC curves: Model-wise performance

Fig5.3.5: Top biomarkers impacting PCOS



5.4 Interpretation and Analysis

The interpretation and analysis stage translated the computational outputs of the models into meaningful clinical and biological insights relevant to Polycystic Ovary Syndrome (PCOS). This phase critically examined model performance, feature significance, and cluster patterns to validate the predictive capacity and interpretability of the proposed diagnostic pipeline.

The **unsupervised clustering analysis** using KMeans and PCA revealed distinct patient subgroups based on biochemical and physiological variables such as BMI, AMH (Anti-Müllerian Hormone), and follicle counts. These clusters suggested inherent heterogeneity among PCOS patients, indicating possible variation in symptom severity and metabolic conditions. For example, one cluster showed elevated BMI and AMH values, correlating with higher PCOS risk, while another reflected more balanced profiles indicative of lower disease probability. The PCA visualizations provided an interpretable two-dimensional perspective, helping to understand how different features contributed to the grouping and how patients distributed across the risk spectrum.

The **SVM decision boundary visualization** offered a practical diagnostic interpretation — clearly separating PCOS and non-PCOS classes based on BMI and AMH levels. Support vectors identified borderline cases within the margin area, representing ambiguous patient profiles that could benefit from additional clinical testing. This visual decision boundary reinforced how subtle physiological differences can significantly influence diagnostic outcomes and underscored the model's sensitivity to hormonal predictors.

In the realm of **supervised learning**, the comparative performance analysis demonstrated that ensemble modeling — combining Logistic Regression, Random Forest, and XGBoost — yielded the highest ROC-AUC values, confirming its strong generalization capability. Individual models revealed different strengths: Logistic Regression provided transparency in feature contribution, Random Forest captured complex non-linear relationships, and XGBoost achieved high predictive accuracy through gradient optimization. The convergence of these models in the ensemble produced a robust consensus classification system capable of maintaining high diagnostic reliability across varying patient profiles.

Feature importance analyses further deepened the interpretation. Both XGBoost and Random Forest highlighted **BMI, AMH levels, RBS (Random Blood Sugar), and follicle counts** as principal determinants of PCOS risk. High AMH levels and increased follicle counts were associated with impaired ovulation, while elevated BMI aligned with metabolic dysregulation — a hallmark of PCOS. These insights confirm clinical understandings from existing medical literature, adding computational evidence to established diagnostic biomarkers.

Lastly, the **ROC curve comparison** indicated that the ensemble model consistently outperformed single classifiers in identifying true positives while minimizing false positives, suggesting its viability as a screening support tool. The results validated that machine learning could effectively complement clinical judgment by revealing multivariate relationships often overlooked in traditional assessment methods.

Overall, the interpretation and analysis confirmed that combining clustering, SVM classification, and ensemble learning not only improved predictive performance but also enhanced biological interpretability. The approach bridged computational outputs with

clinical reasoning, establishing a foundation for deploying explainable AI-driven decision systems in PCOS diagnostics.

5.5 Discussion of Key Findings

The findings from this study emphasize that integrating **metabolic and hormonal biomarkers** provides a powerful and clinically meaningful method for PCOS classification. The combination of parameters such as **BMI, AMH, RBS, and follicle counts** led to clear separation between affected and non-affected groups, both visually (through clustering and PCA) and quantitatively (through model performance metrics). This suggests that PCOS diagnosis can be significantly enhanced when metabolic indicators of insulin resistance and hormonal dysfunction are analyzed together rather than in isolation.

Among all classifiers tested, **XGBoost** demonstrated the strongest performance in capturing **nonlinear interactions** among variables. It effectively modeled subtle dependencies between ovarian follicle counts, AMH levels, and metabolic factors — relationships that traditional linear methods might miss. For instance, XGBoost recognized that abnormal follicle numbers interact with hormonal imbalances differently across ranges of BMI and glucose levels, which aligns with clinical observations of PCOS heterogeneity. Its feature importance plots further highlighted the dominant role of follicular and endocrine parameters in disease prediction, ensuring biologically consistent inference.

Meanwhile, **Logistic Regression** retained substantial interpretability despite its simpler structure. It validated that features like AMH and BMI are statistically significant predictors of PCOS presence, providing directly interpretable coefficients that clinicians can easily understand and trust. This reinforced one of the project's central goals — to balance predictive accuracy with explainability, ensuring that results remain actionable in healthcare contexts.

The ensemble approach, which merged XGBoost, Random Forest, and Logistic Regression, achieved the best diagnostic performance overall, reflecting how integrating different learning paradigms can leverage both interpretability and complexity. Collectively, these findings confirm that combining **machine learning precision with clinical insight** can yield robust and generalizable systems for PCOS diagnosis and prognosis.

5.6 Limitations of Results

The main limitations of this study include the moderate and region-specific dataset, which restricts the generalizability of results across different populations. Hormonal variations due to menstrual cycle phases may have introduced inconsistencies affecting clustering accuracy. The ensemble model, while effective, relies heavily on parameter tuning and may perform differently on external data without recalibration. Additionally, the absence of longitudinal data limits understanding of PCOS progression, and the focus on clinical and biochemical parameters excludes potentially valuable genomic or imaging features. Future work should address these gaps through larger, standardized, and multi-center studies.

6. CONCLUSION AND FUTURE WORK

6.1 Summary of the Research

This study developed an end-to-end, reproducible machine learning framework for the diagnosis and analysis of Polycystic Ovary Syndrome (PCOS). The workflow began with comprehensive data preprocessing, including cleaning, encoding, and normalization of clinical and biochemical data. Using unsupervised clustering (KMeans and PCA visualization), the pipeline provided exploratory structure and feature grouping insights into patient heterogeneity. Then, supervised models — Support Vector Machines (SVM), Random Forest (RF), XGBoost (XGB), and Logistic Regression (LR) — were trained and optimized using grid search cross-validation. Finally, the models were compared through ROC-AUC analysis and integrated into an ensemble classifier, improving PCOS classification performance. The approach successfully unified exploratory clustering, feature interpretability, and predictive modelling, allowing both clinical correlation and robust prediction. Visualizations like SVM decision boundaries and feature importances illustrated which biomarkers (e.g., BMI, AMH, follicle counts) most strongly influenced model decisions.

6.2 Conclusions Drawn from the Results

The ensemble classifier combining Logistic Regression, Random Forest, and XGBoost achieved the highest AUC among all models, showing superior diagnostic accuracy and reliability. Ensemble learning yielded better generalization on unseen data, minimizing model bias and overfitting. The machine learning framework demonstrated that non-invasive clinical markers, such as body mass index (BMI) and Anti-Müllerian hormone (AMH), can effectively indicate PCOS presence, reducing dependency on more invasive or costly procedures. Moreover, the feature visualization and clustering analysis exposed subgroups within the PCOS population — potentially reflecting different disease phenotypes or severity spectra. This provides a foundation for personalized interventions and precision healthcare approaches.

6.3 Contributions of the Study

This study presents a reproducible pipeline design that offers a step-by-step framework encompassing data preprocessing, clustering, classification, visualization, and evaluation, ensuring transparent and systematic workflow execution. By adopting a hybrid modeling strategy that integrates both supervised and unsupervised learning, the framework enhances diagnostic robustness while providing deeper interpretative insights into patient data. A strong emphasis was placed on model interpretability, with feature importance visualizations derived from algorithms such as XGBoost and Random Forest, enabling clinicians to clearly understand the influence of specific biological markers on prediction outcomes. The study also carried out comparative performance benchmarking by systematically evaluating multiple machine learning algorithms using ROC-AUC metrics to identify the most accurate approaches. Ultimately, clinical insight generation was a core achievement, with key variables like BMI, AMH levels, and follicle counts being identified as reliable indicators for targeted early screening in PCOS risk assessment.

6.4 Future Work

The future work of this study focuses on expanding the PCOS diagnostic framework into a more integrated, explainable, and real-time clinical support system. Future research should incorporate hormonal cycle phase corrections to account for natural biological fluctuations and enhance normalization across patients. A major extension will involve the inclusion of multi-omics datasets such as gene expression, proteomic, and metabolomic data to uncover deeper causal biomarkers and molecular signatures of PCOS. Implementing explainable AI tools like SHAP or LIME can improve model transparency, thereby increasing clinician trust in predictive outcomes. Additionally, longitudinal patient data could be used to develop temporal models that capture disease progression and treatment effects over time. Finally, the entire system could be deployed as a lightweight web-based diagnostic platform, enabling healthcare professionals to input patient data and receive real-time, interpretable risk predictions, fostering early detection and personalized management of PCOS.